AFFICIONADO COFFEE ROASTERS

Sales Trend and Time-Based Performance Analysis

A Research Paper and Interactive Dashboard Project

Project Type	Data Analytics / Business Intelligence
Technologies	Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit
Dataset	Afficionado Coffee Roasters sales dataset (Excel)
Dashboard	Interactive Streamlit Web Dashboard
Repository	GitHub – Afficionado-Coffee-Roasters-Analysis

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Abstract

This research paper presents a data analytics project developed for the analysis of sales transactions from Afficionado Coffee Roasters. The study focuses on converting raw transaction-level coffee sales data into meaningful business insights through data preparation, exploratory analysis, visualization, and interactive dashboard development. Python was used as the primary analytical environment, with Pandas and NumPy supporting data manipulation and numerical operations, while Matplotlib, Seaborn, and Plotly were used for visualization. An interactive Streamlit dashboard was developed to present key performance indicators, product performance, store performance, and time-based sales patterns. The project demonstrates how a reproducible Python workflow can transform spreadsheet-based data into a shareable analytical application. The completed dashboard is deployed through Streamlit Community Cloud and the project source files are maintained in GitHub.

Keywords

Coffee Sales Analysis; Data Analytics; Python; Pandas; Streamlit; Data Visualization; Business Intelligence; Sales Trend Analysis; Time-Based Analysis

1. Introduction

Coffee retail businesses generate transaction data that can be used to understand customer demand, product performance, store activity, and sales timing. However, raw transaction records are not immediately useful for decision-making unless they are systematically cleaned, transformed, analyzed, and communicated.

This project develops a complete analytical workflow for the Afficionado Coffee Roasters dataset. The workflow begins with an Excel-based dataset and proceeds through data validation, preparation, exploratory analysis, visualization, and dashboard deployment. The final dashboard allows users to interact with the analysis instead of relying only on static charts.

Pandas is well suited to this type of tabular workflow because it provides DataFrame structures, missing-data handling, grouping, reshaping, and Excel input/output capabilities. These features support a repeatable process from raw spreadsheet data to analytical results.

2. Problem Statement

The main problem addressed by this project is the difficulty of extracting actionable sales information from raw coffee transaction data. Without a structured analytical process, it can be difficult to determine which products and stores perform strongly, which periods of the day experience higher transaction activity, and where management attention may be required.

Therefore, the project aims to create a reliable and interactive analytical solution that converts transaction records into visual business information and makes the results accessible through a web-based dashboard.

3. Objectives

The project objectives are to understand the structure and quality of the supplied sales dataset; validate missing values, duplicate records, data types, and relevant transaction fields; prepare the dataset using Python and Pandas; analyze sales performance by product and store; analyze transaction activity by hour; create meaningful KPIs; develop interactive visualizations using Plotly and Streamlit; deploy the completed dashboard; and document the workflow for reproducibility and portfolio presentation.

4. Dataset Description

The project uses the supplied Afficionado Coffee Roasters Excel workbook. The workbook contains transaction-oriented coffee sales information. The analytical workflow is designed around the available fields in the supplied dataset, including transaction time, product-related information, store-related information, and sales/quantity measures.

The analysis intentionally uses only fields that are present in the supplied workbook. In particular, the available transaction-time information supports hour-based analysis. A calendar date or day-of-week field should not be assumed unless it is actually present in the source workbook.

5. Research Methodology

The project follows a structured data analytics lifecycle consisting of data acquisition, data quality assessment, data preparation, exploratory analysis, visualization, dashboard development, and deployment.

Data acquisition: The supplied Excel workbook was loaded into a Pandas DataFrame.

Data quality assessment: The dataset was inspected for shape, column names, missing values, duplicate records, and data types.

Data preparation: Relevant fields were converted to suitable data types. Transaction-time information was prepared for hour-based analysis, and calculated measures were created where required by the dashboard.

Exploratory analysis: Group-based aggregations were used to compare sales or quantity across products, stores, and transaction hours.

Visualization: Charts were designed to communicate comparisons and time-based patterns. Plotly was used for interactive visual exploration.

Dashboard development: Streamlit was used to combine KPIs, charts, filters, and explanatory text into a single web application.

Deployment: The application was connected to GitHub and deployed through Streamlit Community Cloud, producing a shareable live URL.

6. Technologies Used

Python was used as the primary programming language. Pandas supported data loading, cleaning, transformation, aggregation, and analysis. NumPy supported numerical operations. Matplotlib and Seaborn supported exploratory visualization. Plotly provided interactive charts. Streamlit was used to create the web dashboard. Microsoft Excel was the source format for the dataset, GitHub was used for version control and project hosting, and Streamlit Community Cloud was used for deployment.

7. Analytical Framework

Key Performance Indicators: The dashboard summarizes the dataset using KPI cards such as transaction count, quantity sold, revenue/sales measures, and average performance measures where the corresponding fields are available.

Product Performance Analysis: Product-level aggregation is used to compare the contribution of different coffee products and identify high-volume or high-performing products.

Store Performance Analysis: Store-level analysis compares transaction activity and sales-related measures across store locations.

Time-Based Analysis: Transaction time is transformed into an hour-based analytical dimension. Hourly aggregation helps identify periods of relatively high or low transaction activity, supporting staffing, inventory readiness, and promotional planning.

8. Dashboard Design

The Streamlit dashboard is designed as a business-facing analytical interface rather than a collection of raw Python outputs. The interface combines a clear project title, project description, KPI cards, interactive charts, filters, and explanatory notes. Users can explore the data without editing the underlying Python code.

The dashboard includes an executive KPI summary, product performance comparison, store performance comparison, hourly or peak-time analysis, interactive filters, and supporting explanatory information.

9. Findings and Discussion

The project demonstrates that transaction-level coffee sales data can be transformed into a practical business intelligence interface. Product-level aggregation makes it easier to identify high-performing and lower-performing products. Store-level comparisons provide a direct view of differences in transaction activity between locations. Hourly analysis reveals periods of higher and lower transaction activity and can support operational planning. KPI cards provide an immediate summary of overall performance before detailed investigation. Interactive visualization improves usability by allowing users to explore the dataset from different perspectives.

These findings are presented qualitatively to avoid inventing numeric results. Exact numeric findings should be taken directly from the final validated dataset and deployed dashboard.

10. Business Implications

The analytical framework can support several operational decisions. Peak-hour information can help managers plan staffing and service capacity. Product performance can inform inventory and promotional decisions. Store comparisons can help identify locations that perform differently from the overall business pattern. Together, these views provide a foundation for evidence-based sales and operational planning.

11. Limitations

The analysis is limited to the fields and time information available in the supplied workbook. If the source dataset does not contain calendar dates, true daily or monthly trend analysis cannot be reliably inferred. The project is descriptive and does not attempt to forecast future sales. The dashboard does not establish causal relationships between sales and external factors such as weather, promotions, holidays, or customer demographics unless such variables exist in the dataset. Dashboard conclusions depend on the quality and completeness of the supplied source data.

12. Future Scope

Future work can include daily, weekly, and monthly sales trends if calendar dates become available; sales forecasting using time-series or machine-learning models; customer segmentation if customer-level data becomes available; inventory and stock-out analysis; promotion, holiday, weather, and seasonality variables; automated data refresh; and role-based dashboard views for management, store managers, and analysts.

13. Conclusion

The Afficionado Coffee Roasters Sales Analysis project demonstrates an end-to-end data analytics workflow from spreadsheet data to a deployed interactive dashboard. Python and Pandas provide the analytical foundation, while visualization libraries transform processed data into understandable charts. Streamlit extends the analysis into an interactive web application that can be shared through a live URL.

The project provides practical evidence of skills in data cleaning, exploratory analysis, business-oriented visualization, dashboard development, and deployment. It also illustrates the value of maintaining data, code, dependencies, and deployment configuration together in a GitHub repository.

14. References

1. Pandas Documentation. Python Data Analysis Library. Official documentation, accessed 2026.
2. Streamlit Documentation. Deployment and Streamlit Community Cloud documentation, accessed 2026.
3. NumPy Documentation. Official documentation, accessed 2026.
4. Plotly Python Documentation. Official documentation, accessed 2026.
5. Matplotlib Documentation. Official documentation, accessed 2026.

Appendix – Project Links

GitHub Repository: https://github.com/mrklavanya0108-stack/Afficionado-Coffee-Roasters-Analysis

Live Dashboard: https://afficionado-coffee-roasters-analysis-xky6d3pdlh8wkrkrqbmq6s.streamlit.app/